Comparative Study of FTO and Carbon Cloth for Electrocatalytic Hydrogenation of Phenol on Platinum Catalyst

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The scientific community is seeking renewable resources as an alternative to fossil fuels due to their widespread use and environmental damage. Thus, converting lignocellulose biomass (LCB) into value-added bioproducts is environmentally beneficial and sustainable. LCB is mostly hemi(cellulose) and lignin. Lignin, the principal aromatic bio-source in LCB [1]. Lignin was converted into value-added chemicals by acid/base catalysis, catalytic reduction, (hydro)thermal treatment [2]. High temperature and H_2 pressure are needed for thermal catalysis to upgrade high oxygen bio-oil. An alternative approach is electrocatalytic hydrogenation (ECH), which is beneficial due to in situ generation of hydrogenation (Hads) from water splitting reaction and then that hydrogen is used for the hydrogenation of lignin model compounds. Moreover, the ECH operates at milder reaction conditions with no catalyst deactivation and coke formation. In this context, lignin-derived ECH typically uses noble metal-based catalysts [3]. This study is devoted to examine the different substrate effects for ECH of simplest lignin model compound i.e., phenol.

The ECH of phenol was performed by employing commercial 5 wt.% PtC catalyst. The phenol ECH was conducted by using three electrode H-cell i.e., working electrode (PtC coated FTO & carbon cloth) Ag/AgCl (reference electrode) and Pt wire (counter electrode) equipped with PEM. In a standard experiment, 20 mM phenol in 0.2 M $HClO_4$ was used at -2.5 V (for FTO) and -0.7 V (for carbon cloth) versus Ag/AgCl for 6 h at ambient temperature. The obtained products were identified and quantified via GC-MS and FID.

Initially, utilizing catalyst coated FTO (3 mg/cm^2) as working electrode and 10 mM of phenol as substrate, approximately 90% conversion was observed after 6 hrs with high

selectivity towards cyclohexanol more than 90% as primary product while cyclohexanone was less than 10% as a secondary product **(fig 1a)**. Then, with the increase in substrate concentration from 10 mM to 20 mM, the conversion further improves to more than 90% while same product distribution was observed with cyclohexanol as primary product more than 95% selectivity and cyclohexanone as secondary product with less than 5% selectivity **(fig 1b)**. It is evident from this data that the reaction followed the consecutive pathway as some of the phenol molecules were directly converted to cyclohexanol by utilizing 6 Hads while some molecules were converted into cyclohexanone by taking up 4 Hads. Additionally, cyclohexanone selectivity drops with time while cyclohexanol selectivity increases which means that cyclohexanone was further hydrogenated to cyclohexanol by taking 2 more Hads.

Using carbon cloth instead of FTO for 20 mM phenol with same amount of catalyst loading (3 mg/cm^2) resulted in nearly 80% conversion after 6 hours, with cyclohexanol as the main product (~90% selectivity) and cyclohexanone as the minor product (~10% selectivity) **(fig 1c)**. These results are consistent with the results on FTO.

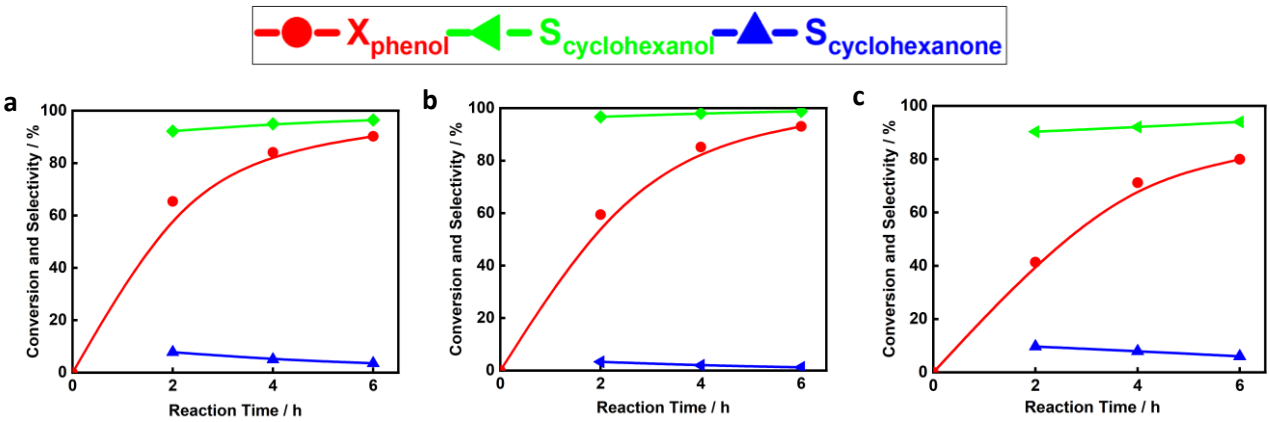

Fig 1. Phenol conversion profile with product selectivities of cyclohexanol & cyclohexanone (a & b) 10 & 20 mM of phenol on FTO coated catalyst (c) 20 mM of phenol on carbon cloth impregnated catalyst.

In summary, carbon cloth impregnated catalyst has significant catalytic activity at lower potential (-0.7 V) but a bit inferior selectivity towards targeted product than FTO coated catalyst, which retains selectivity at overpotential (-2.5 V). So, more energy was consumed by FTO supported catalyst than carbon cloth supported catalyst.

References

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